

Sessional 2 solution

Question No 1:

Determine the truth value of each of these statements if the domain for all variables consists of all integers.

a) $\forall n \exists m (n^2 < m)$

The import of this statement is that no matter how large n might be, we can always find an integer m bigger than n^2 . This is certainly true; for example, we could always take $m = n^2 + 1$.

b) $\exists n \forall m (n < m^2)$

This statement is asserting that there is an n that is smaller than the square of every integer; note that n is not allowed to depend on m , since the existential quantifier comes first. This statement is true, since we could take, for instance, $n = -3$, and then n would be less than every square, since squares are always greater than or equal to 0.

c) $\exists n \exists m (n + m = 4 \wedge n - m = 1)$

The statement is that the system of equations $\{n + m = 4, n - m = 1\}$ has a solution over the integers. By algebra we see that there is a unique solution to this system, namely $n = 2.5$, $m = 1.5$. Since there do not exist integers that make the equations true, the statement is false.

d) $\exists n \exists m (n^2 + m^2 = 5)$

The statement is that the equation $n^2 + m^2 = 5$ has a solution over the integers. This is true; in fact there are eight solutions, namely $n = \pm 1$, $m = \pm 2$, and vice versa.

Question No 2:

If p is a prime number, must $2^p - 1$ also be prime? Prove or give a counterexample?

Statement: If p is a prime number, then $2^p - 1$ is also a prime number.

We suspect that $2^p - 1$ is not a prime number for all primes p . However, we can only be sure by finding a counterexample.

Counterexample Let $p = 11$. Note that 11 is a prime.

$$2^p - 1 = 2^{11} - 1 = 2048 - 1 = 2047 = 23 \cdot 89$$

Since 23 and 89 are both not equal to 1, $2^p - 1$ is not a prime (by the definition of prime).

Question No 3 :

For any positive integer m, if m is prime then $m^2 + 4$ is also prime.

Counterexample:

Assume $m = 2$,

Then it follows that:

$$m^2 + 4 = (2)^2 + 4$$

$$= 4 + 4$$

$$= 8$$

As 8 is a composite number and not a prime, this is a contradiction, and the original claim is false.

Prove: If m and n are both perfect squares, then nm is also a perfect square.

We assume that m and n are both perfect squares. By the definition of a perfect square, $m = s^2$, and $n = t^2$, where both s and t are some integers.

$$mn = s^2 t^2$$

$$mn = (st)^2$$

$$mn = h^2, \text{ where } h = st \text{ and an integer}$$

Therefore, mn is a perfect square

Question No 4 :

Rearrange and re-write the following premises to make them consistent with the conclusion.

P1: If a pet is above all the dogs, then it is above all the animals.

P2: If a pet is not above all the cats, then it is not a panther.

P3: Every black pet is a panther.

P4: Any pet that is above all the cats in the household is above all the dogs.

Conclusion: If a pet is black, then it is above all the animals

Solution:

First step: Re-write every sentence as an implication.

P1: If a pet is above all the dogs then it is above all the animals.

P2: If a pet is not above all the cats then it is not a panther

P3: If a pet is black then it is a panther.

P4: If a pet is above all the dogs then it is above all the animals.

2nd step: Re-order and re-write the sentences to ensure transitivity.

P1: If a pet is black then it is a panther.

P2: If a pet is a panther then the pet is above all the cats.

P3: If a pet is above all the cats then it is above all the dogs.

P4: If a pet is above all the dogs then it is above all the animals.

Conclusion:

If a pet is black, then it is above all the animals.

Question No 5 :

For each of the following parts, determine if the given relations are reflexive, symmetric, or transitive. Indicate whether the statements are True or False, and support your answers with examples.

Part I- Determine if the following relations are reflexive or not.

(1) $(\forall x \in \mathbb{R}) (x = x).$

(2) $(\forall x \in \mathbb{R}) (x < x).$

(3) $(\forall a \in \mathbb{N}) (a \mid a).$

Part II- Determine if the following relations are symmetric or not.

(1) $(\forall x, y \in \mathbb{R}) ((x = y) \rightarrow (y = x)).$

(2) $(\forall x, y \in \mathbb{R}) (x \geq y \rightarrow y \geq x).$

Part III- Determine if the following relations are transitive or not.

(1) $(\forall x, y, z \in \mathbb{R}) (((x = y) \wedge (y = z)) \rightarrow (x = z)).$

(2) $(\forall x, y, z \in \mathbb{R}) ((x < y \wedge y < z) \rightarrow (x < z)).$

Solution:

Part I- Determine if following relations are reflexive or not.

(1) $(\forall x \in \mathbb{R}) (x = x)$.

True. Example: $x = x$ will always be reflexive, ($1=1$, $3=3$, $4.51 = 4.51$, etc.)

(2) $(\forall x \in \mathbb{R}) (x < x)$.

False. Example: $x < x$ will never be reflexive. ($0.5 < 0.5$? Incorrect.)

(3) $(\forall a \in \mathbb{N}) (a \mid a)$.

True. An integer can always divide itself. ($4/4$, $5/5$, $7/7$, etc.)

Part II- Determine if the following relations are symmetric or not.

(1) $(\forall x, y \in \mathbb{R}) ((x = y) \rightarrow (y = x))$.

True. Example: $x = 5$, $y = x$ that means $y = 5$, then $y = x$ which is true.

(2) $(\forall x, y \in \mathbb{R}) (x \geq y \rightarrow y \geq x)$.

False. Example: $x = 8$, $y = 7$ then $y \geq x$? $7 \geq 8$? Incorrect. It will only be true when x and y will be equal.

Part III- Determine if the following relations are transitive or not.

(1) $(\forall x, y, z \in \mathbb{R}) (((x = y) \wedge (y = z)) \rightarrow (x = z))$.

True. Example: $x = 5$, $y = x$, $y = 5$ and $y = z$ meaning $z = 5$ then $x = z$ meaning $5 = 5$ and that is true.

(2) $(\forall x, y, z \in \mathbb{R}) ((x < y \wedge y < z) \rightarrow (x < z))$.

True. Example: $x = 4$, $y = 5$ and $y < z$ then $z = 6$. Then $x < z$ and $4 < 6$ and that is correct.